Networks I: Concepts and Algorithms

Winter 2026 – Pau Milán

Problem Set 3

Due on *Sunday, March 15*

INSTRUCTIONS: *You may work in groups of* ***at most four****. Please upload your scanned responses to* ***google classroom*** *by the deadline. All files should be contained in* ***a single zip file****. The name of the file should be "PS3-names" where "names" correspond to your surname(s). Each script file should begin with preamble that contains your name(s), date, and briefly states what you are doing. Please* ***comment extensively*** *your steps in the code. Codes will be tested and must* ***run without errors****. Besides providing the scripts, you should answer all questions neatly in a* ***separate pdf file*** *providing the plots and figures that are required to make your point.*

Cascades and Network Structure

1. In this question, you will explore via a numerical simulation the impact that community structure has on the dynamics of disease.

   (a) The "planted partition" model we saw in class is a one-parameter version of the stochastic block model that generates simple synthetic networks with community structure of varying strength. Generate and create simple visualizations of three graphs, for $n = 100$ (population size), $c = 2$ (number of communities), $d = 10$ (average degree), and $\varepsilon = \{0, 4, 8\}$ (strength of communities as shown in the slides).

   Recall that in the slides we saw how ε maps into the probabilities p (within-group) and q (across-group). Comment on the strength of the community structure each of these graphs exhibits.

   (b) A simple SI spreading process is one in which the probability that a vertex in the infected state I transmits its infection to a particular uninfected neighbor is a constant p, and we evaluate that transmission possibility at most once for that edge over the lifetime of the epidemic. Initially, at time $t = 0$, all vertices are in the uninfected state S ("susceptible"), time proceeds in discrete steps and vertex state updates are applied simultaneously when moving from $t \to t + 1$. A simulated epidemic is deemed complete when no new infected nodes are produced in a time step. The epidemic's size s is the fraction of nodes in the I state when the epidemic is complete, and its length ℓ is the

time $t = \ell$ at which the last node in the epidemic became infected. To begin the epidemic, at time $t = 1$, choose a node uniformly at random to infect.

- Using the planted partition model from part (a), you can generate any number of synthetic networks to study the behavior of the simple Independent Cascade model in the slides. Using $n = 1000$, $d = 8$, and $\varepsilon = 0$ measure the average epidemic size $\langle s/n \rangle$ and average epidemic length $\langle \ell \rangle$ as a function of $p \in [0, 1]$ (the independent probability that an infected individual transmits to each neighbor). Make two figures, showing these measured relationships.

  Hint: To get good figures, you will want smooth functions, which means averaging the measured s and ℓ over multiple draws from your data generating process, which in this case is the network generator and the simulation. You will also want to vary p slowly enough over the [0, 1] interval to get good resolution on how the average epidemic size changes, especially in the regions of p where s/n or ℓ are changing quickly.

- Comment on the qualitative behavior of these measured relationships as a function of the transmission probability p, and discuss your results relative to your expectations. Is there a critical value of p where a phase transition occurs?

(c) Now investigate whether the strength of community structure enhances, limits, or has no effect on epidemic size. Let $n = 200$ and $d = 8$, and consider various combinations of the two parameters: $p \in [0, 1]$ and $\epsilon \in [0, 2c]$.

- Present your characterization of how community structure strength ϵ impacts epidemic size s/n and/or epidemic length ℓ using one or more figures that show the relationship clearly (smooth functions).
- Discuss the qualitative shape of these functions, and how they contrast, if at all, with your results from part (1 b).
- Provide a brief intuitive explanation for why community structure strength does or does not impact the shape of the epidemic.

Optimal Seeding

2. In this question you will simulate a couple of diffusion models and find the optimal seeds using real-world network data that you can freely choose. I recommend the following website that has a large compendium of networks – to minimize computational burden I recommend you choose a small(ish) network.

   Consider an adoption threshold τ_i which determines the minimum number of neighbors that must adopt such that a farmer decides to adopt a new planting technology.

   (a) For $\lambda = 1$ and $\lambda = 2$, conduct technology diffusion simulations for your village, where each individual draws an adoption threshold τ from a $\mathbb{N}(\lambda, 0.5)$, but truncated to be strictly positive.

   HINT: In the simulations, when a node is connected to at least τ individuals who are informed, she becomes informed in the next period. Run the model for four periods.

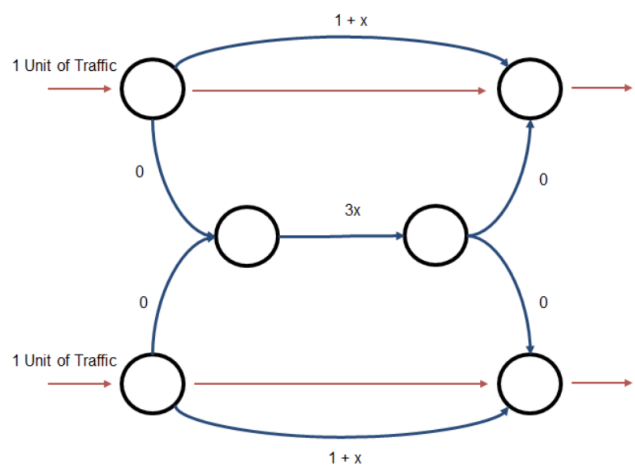
Given the randomness built into the model, simulate the model 60 times **for each potential pair of seeds**, and create a measure of the average information rate induced by each pair

(b) Provide the identity of the optimal pair of seeds under $\lambda = 1$ and $\lambda = 2$

(c) Now run the same Monte Carlo simulation as in part (a) but using randomly chosen seeds. Compare the rates of information diffusion with the optimal seeds in part (a) and discuss the results.

Routing on Roads

3. Consider the traffic flow game pictured in the figure below. There are two origin-destination pairs. A unit of traffic needs to flow from the upper left to the upper right, and another unit needs to flow from the lower left to the lower right. The cost functions are given in the figure.

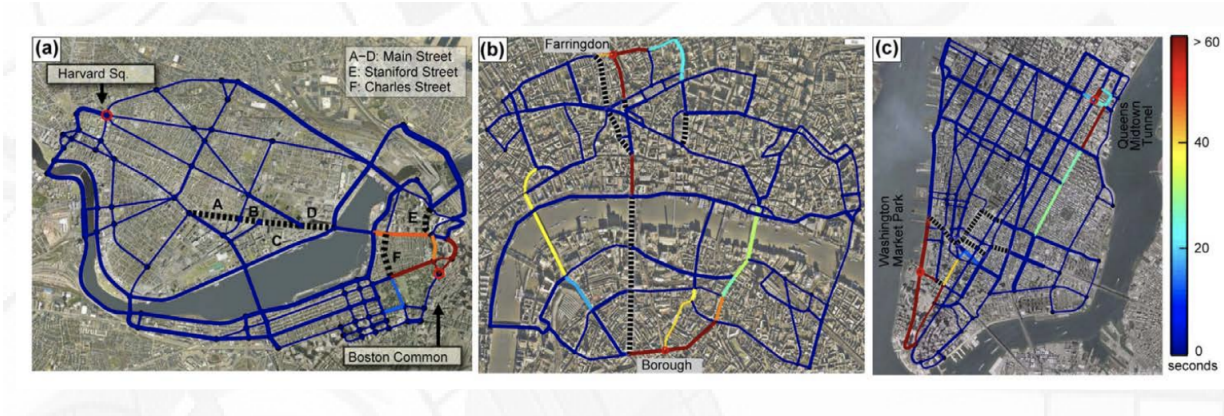


(a) What is the socially optimal routing, and what is its total cost?

(b) What is the equilibrium routing? What is the welfare loss relative to the optimum?

(c) Suppose you can impose constant tolls on some edges and constant subsidies on others. Design a system of tolls and subsidies to implement the social optimum as an equilibrium.

4. (*Extra points*) Let's try and replicate the paper by Youn et al. (2008) we saw in class that identified roads that should be closed in real road networks, as in the figure below that I showed you in class



The paper assumes a latency function given by

$$l_{ij} = \frac{d_{ij}}{v_{ij}} \left[1 + \alpha \left(\frac{x_{ij}}{p_{ij}} \right)^{\beta} \right]$$

where d_{ij} is the distance of the link between i j, v_{ij} the speed limit (assume 60 kph on all links for simplicity), x_{ij} the flow, and p_{ij} the capacity of the road segment. The parameters α and β have been fitted to empirical data as $\alpha = 0.2$ and $\beta = 10$

(*Hint: Notice that here we are indexing links by the two vertices it connects, whereas in the notes we just indexed a link directly – so in the notes we say the flow in link i is x_i and here we say the flow in the link between i and j is x_{ij}. This is just to keep the notation as close as possible to Youn et al. (2008)*)

(a) Go to this address and download one of the available road networks (they even have Barcelona!). You will see that the data contains all the information needed to compute the latency above. Import the data and plot your network. (*Hint:* if the network is too large, take a random subset of it, as it will otherwise make your code too sluggish)

(b) Compute the equilibrium (i.e. Nash) flow of your network by minimizing the potential function that we saw in class.

(c) Compare the costs of the Nash flow on the original network with those on networks where one of the streets is closed to traffic. Repeat this by closing each street (one at a time).

(d) Can you find street(s) where closing it decreased the cost of the Nash flow? If you did, you have found one of the black streets in the figure above. You may now call the mayor and let her know the good news.

Opinion Dynamics

5. Download the Facebook100 data, containing facebook networks for 100 universities in the US. We want to analyze how the different networks generate different opinion dynamics. Keep the largest component of each network only.

(a) To simplify things we will assume that each node "listens" to all her neighbors (and herself) with the same probability. In other words, we assume that $T_{ij} = 1/(1 + d_i)$ for all $j \in N_i$ and also $T_{ii} = 1/(1 + d_i)$.

(b) Since we don't have any way of performing sentiment analysis on this data, we will instead simulate innate/original opinions from a uniform(0,1). In other words, assign to each node a $p_i(0)$ that is drawn from a uniform random variable

(c) Since we have only one component and all edges are bi-directional (and we assume everyone pays some attention to themselves) then we should obtain convergence to consensus. Compute the influence vector for each network and report a histogram of the influences obtained across all vectors. What is the maximum influence? and the median influence?

(d) Plot a histogram of the consensus opinions across all networks. Which is the most popular (i.e. modal) opinion?

(e) Now we want to test the speed of learning. For each network, calculate the first period, $t^\star$, where $\left(\sum_i \tilde{d}_i(A)\left(p_i(t) - p_i(\infty)\right)^2\right)^{\frac{1}{2}} < \varepsilon$ for $\epsilon = 0.1$ and $\tilde{d}_i(A) = \frac{d_i(A)}{\sum_i d_i(A)}$.

(f) Construct a plot with the second eigenvalue of the network on the x-axis and $t^\star$ on the y-axis. Do you get the relationship you expected? What happens as you lower the value of ϵ?

(g) (*Extra points*) Now assume that the $p_i(0)$ that you obtained from the uniform distribution above is in fact the innate opinion s_i of agent i. Simulate a susceptibility parameter α_i also from the Uniform (0,1) for each agent i and solve, for each network, the steady-state value $p^\star$ under the Friedkin-Johnsen model. Construct a histogram of $p^\star$ across all networks. What is the most popular opinion? How does your answer change of α_i is distributed Uniform(0.5, 1)?

(h) (*Extra points*) Pick your favorite network from Facebook100 and find the optimal vector $\alpha^\star$ that maximizes the sum of opinions $\mathbf{1}'p^\star$ as in Abebe et al. (2018).